

[CH3] Babbage's Dream Comes True

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- Babbage placed the blame for the failure of his calculating engines upon the British government and the fact that they refused to offer him more funding
- Of the two machines (The analytical engine and the difference engine), the difference engine was less interesting but it came closest to being built
 - He worked on it for about a decade (starting in the early 1820s and culminating in 1833)
 - Ada Byron was impressed by it. Her mathematical studies were guided by Babbage and she later wrote the best 19th century account of the analytical engine
- The difference engine was never fully built, because Babbage abandoned it for a new invention: the Analytical Engine
 - While the difference engine could only produce tables, the analytical engine would be capable of any mathematical computation
 - He wanted to eliminate the need for human intervention by feeding the results of the computation back into the machine
 - ★ ◦ The most significant concept was the separation of arithmetic and numbers storage
 - Organizing the calculation of the analytical engine (basically programming it) was a challenge for Babbage for several years
 - He then discovered the Jacquard loom. It was a device that took instructions in the form of punched cards and could offer infinite combinations of pattern weaves
 - After losing government funding, no attempt at building the analytical engine was ever made. It became a paper project and the engine became an end in itself.
 - Where the difference engine had the goal of creating tables, the analytical engine had no real goal or purpose. *"apart from Babbage himself, almost no one wanted or needed the Analytical Engine"*
 - He was invited to Turin, Italy to give a seminar on his engines, where he convinced Lieutenant Luigi Menabrea to write an account of the Analytical Engine. Menabrea published it in French and later also became prime minister of Italy
 - Ada, who had married the earl of Lovelace, was asked by Babbage to translate the article into English
 - She added her own notes, quadrupling the length of the original text and described programs, the technical base for most of which was Babbage's work.
- Years after Babbage had understood that no funding would come his way for the engines, Georg Scheutz and Edward Scheutz, two Swedes, built a full scale difference engine. It took them nearly 20 years, but it won the gold medal at the 1855 Paris Exposition
 - It was purchased by the Dudley Observatory in Albany, NY
 - The British government ordered one for the computation of life insurance tables
 - The price was low (GBP 1200), but it was unreliable and had to be adjusted frequently.
- Babbage died at the age of 80, without ever having built his engines
- His inventions were ahead of their time. Had he tried in 1890 what he had been trying in 1820, it would have been pretty straightforward.
- His failure discouraged scientists and engineers from trying other digital approaches. Instead, they moved to analog computing

Analog Computing

- Means building a physical model of the system that has to be analyzed, instead of computing with numbers.

Lord Kelvin (William Thomson)

- 1876, British scientist Lord Kelvin (William Thomson), invented a tide-predicting machine
- Because tides involve very complex calculations, tide tables were only made for a few major ports around the world
- ★ • The main problem with analog computing technology is that it is not general-purpose. There is a specific device

for a specific problem.

Vannevar Bush

- Vannevar Bush invented the **Profile Tracer**, a device that could plot ground contours
- 1919 bush became an instructor in electrical engineering at MIT.
- 1924 he built the **Product Integrator**, a device to solve problem in electrical transmission
- 1928-1931 he built the **Differential Analyzer**, a device that could solve differential equation
 - Unlike other analog devices, this one addressed a very wide range of problems
 - It was not a general purpose computer, but it was *the most important computing machine developed between the wars*
- Analog computers only worked when a model of the problem could be built. There are problems for which it is necessary to work with numbers. This is the digital approach

Fry Richardson

- Fry Richardson was a pioneer of numerical meteorology. He believed that the only way to make reliable weather forecasts was mathematically
 - He was able to perform successful calculations, but he was very slow (it took him 6 weeks). He estimated that 32 human computers could keep up with the weather
 - He wrote "**Weather Forecasting by Numerical Process**"
 - When the Air Ministry took over the British Meteorological Office, he resigned. He was unwilling to work directly for the armed forces
 - He never worked on numerical meteorology again

Leslie John Comrie

- Leslie John Comrie had a passion for computing.
- At the British Astronomical Association, he organized a team of 24 human computers to produce astronomical tables
- In 1925 he took up a permanent position with the Nautical Almanac Office
 - Here the process had changed very little.
 - He replaced the freelance computers with clerical work and standard calculating machines
 - He brought punched-card machines (and therefore IBM) into the world of scientific computing
 - 1930 he became the director of the Nautical Almanac Office
 - He was known as the world's leading expert on computing
 - He later became frustrated with the stubborn, change-resistant bureaucracy of the NAO and left to create the **Scientific Computing Service Limited**
 - This was the first for-profit calculating agency

Howard Hathaway Aiken

- Graduate student at Harvard University in the US
- He was a 33-year old electrical engineer when he started research at Harvard
- He needed to solve a nonlinear differential equation, but the current technology could generally only solve ordinary differential equations
- This pushed Aiken to propose a large-scale digital calculator
- He proposed this to the physics department, which was not overly excited
- He found out about Charles Babbage's work and was inspired. He saw himself as Babbage's successor
- He proposed his idea to George Chase, chief engineer at Monroe Calculating Machine Company
- Chase believed in the idea but was unable to secure the support of Monroe. Chase encouraged Aiken to get in touch with IBM
- IBM chief engineer James Bryce understood the importance of the proposal and got 15,000 USD in funding, which was later increased to 100,000 USD
- Aiken's proposal barely constituted a design. It was a general set of functional requirements
- Aiken was essentially a consultant while IBM engineers designed the machine

- In 1943, the first tests were being run in secret on the **Automatic Sequence Controlled Calculator** (also named the **Harvard Mark I**)
- At the revealing ceremony, Aiken completely failed to recognize the contribution that IBM and their engineers had made
- This angered Thomas Watson so much that he decided to build the **Selective Sequence Controlled Calculator**. This was meant to be a better machine in order to prove IBM's superiority and recapture the spotlight

Alan Mathison Turing

- Developments were occurring in mathematical logic at the same time as the developments in computing
- Most of the people involved in building computers were not aware of these developments in mathematical logic, so they had little effect on computer design
- Alan Turing was one of the individuals most closely associated with these developments, along with Alonzo Church
- Both of them were trying to answer the Entscheidungsproblem posed by David Hilbert: *"Did there exist a definite method or process by which all mathematical questions could be decided"*
 - Both Turing and Church answered this
 - Church used conventional mathematics
 - Turing used a conceptual computer, now known as the Turing Machine
- The **Turing Machine** was a thought experiment. It used a scanning head and a potentially infinite length of tape. It could be programmed with an "instruction table". It was universal in the sense that it could compute any function that it was possible to compute. It was computing reduced to its simplest form
- Turing claimed that a universal computer should be able to tackle problems in any tractable area of human knowledge. It embodied **all the logical capabilities of a modern computer**
- He published a paper *"On Computable Numbers with an Application on the Entscheidungsproblem"* in which he described the machine.
 - He showed that not all mathematical questions were decidable and that one could not always determine whether or not a mathematical function was computable
- During WW2 he was part of the Bletchley Park code-breaking operation and used computing machinery to break Nazi codes.
- His work greatly inspired Von Neumann, thereby influencing the invention and design of modern day computers